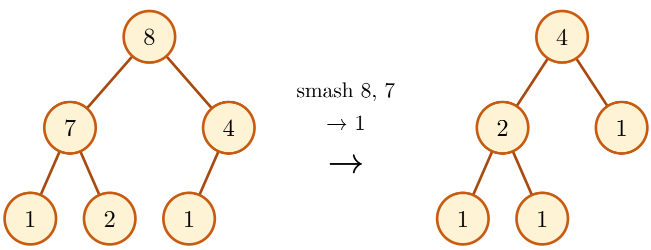
Last Stone Weight

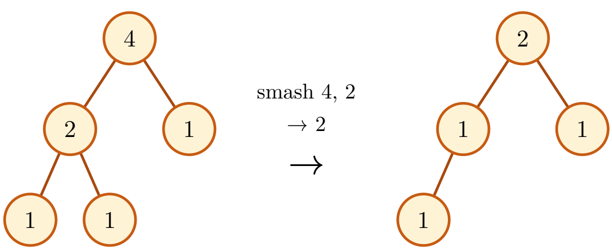
LeetCode 1046 · Easy

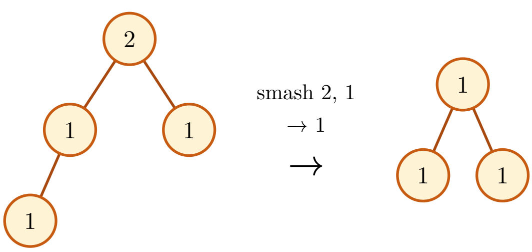
Smash the two heaviest stones together, repeatedly, until at most one is left. If they're equal both are destroyed; otherwise the difference goes back in. A max-heap hands you the two heaviest in $O(\log n)$ each round instead of a full rescan.

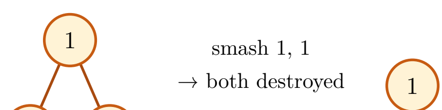
```python
import heapq

def lastStoneWeight(stones):
    heap = [-s for s in stones]
    heapq.heapify(heap)
    while len(heap) > 1:
        y, x = -heapq.heappop(heap), -heapq.heappop(heap)
        if y != x:
            heapq.heappush(heap, -(y - x))
    return -heap[0] if heap else 0
```









One stone remains, weight **1** — matching `lastStoneWeight([2, 7, 4, 1, 8, 1]) == 1`.